

A GUI for Magboltz

IFIN-HH (Romania)
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WG4 — Modeling and Simulations

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Magboltz in Detector Physics

- ▶ Monte Carlo electron transport simulation
- ▶ Fundamental tool for:
 - ▶ Gas detector design
 - ▶ Drift properties
 - ▶ Diffusion
 - ▶ Townsend / attachment
- ▶ Used across:
 - ▶ MPGDs
 - ▶ TPCs
 - ▶ RPC studies
 - ▶ Simulation chains (Garfield++)

Magboltz remains a cornerstone of gas detector modeling.



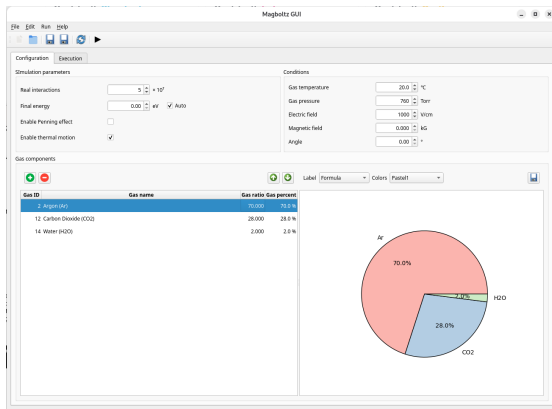
The CLI Workflow Reality

- ▶ Text-based input cards
- ▶ Text-based output parsing
- ▶ Typical friction points:
 - ▶ Input syntax errors
 - ▶ Iterative parameter tuning
 - ▶ Manual data extraction
 - ▶ Ad-hoc plotting scripts
 - ▶ Student onboarding difficulty

Powerful physics engine, but workflow-heavy interface.

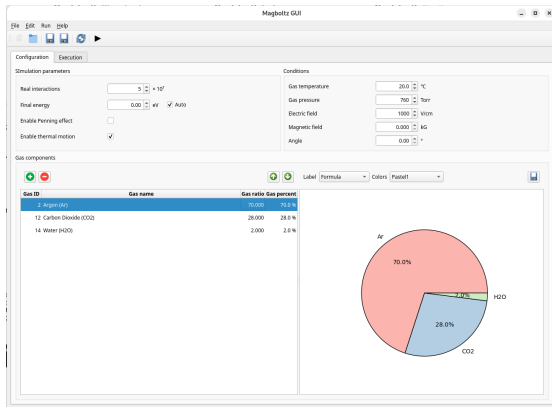
Magboltz GUI

- ▶ Python + PyQt6 application
- ▶ Cross-platform
- ▶ Designed as a workflow layer, not a replacement
- ▶ Core functions:
 - ▶ Input file creation & editing
 - ▶ Magboltz execution wrapper
 - ▶ Output parsing → structured model
 - ▶ Data export → CSV / JSON / XML
 - ▶ Built-in plotting tools



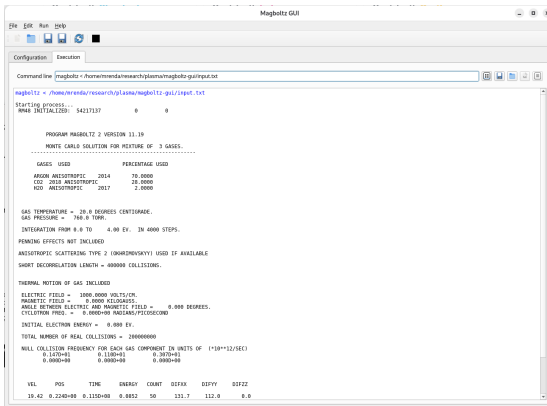
Simulation & Gas Configuration

- ▶ Editable simulation parameters
- ▶ Gas mixture editor
- ▶ Ratio → automatic normalization
- ▶ Visual mixture representation
- ▶ Reduced input mistakes during iteration



Execution Workflow

- ▶ Direct Magboltz invocation
- ▶ Command line transparency
- ▶ Live stdout/stderr streaming
- ▶ Run/Stop controls
- ▶ Raw output persistence
- ▶ Reopen previous runs



```
Magboltz GUI
File Edit Run Help
Configuration Execution

Command line /home/mrenda/research/plasma/magboltz-gui/input.txt

magboltz < /home/mrenda/research/plasma/magboltz-gui/input.txt
Starting process...
NAME INITIAL(2D) 54217137 0 0

PROGRAM MAGBOLTZ 2 VERSION 11.19
MONTH CARLB SOLUTION FOR MIXTURE OF 3 GASES.
-----
GASES USED PERCENTAGE USED
ARGON ANISOTROPIC 2014 70.0000
CO2 2018 ANISOTROPIC 20.0000
N2O ANISOTROPIC 2017 2.0000

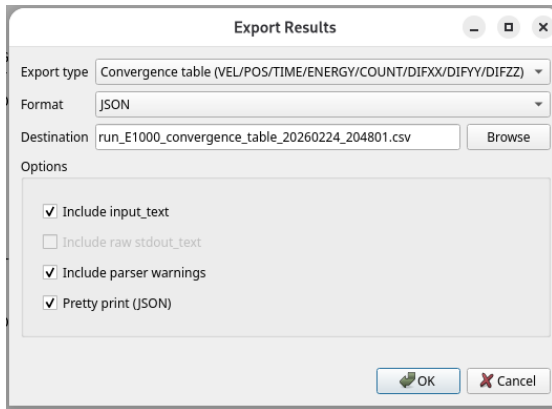
GAS TEMPERATURE = 20.0 DEGREES CENTIGRADE.
GAS PRESSURE = 740.0 TORR.
INTEGRATION FROM 0.0 TO 4.00 EV. IN 4000 STEPS.
PENNING EFFECTS NOT INCLUDED
ANISOTROPIC SCATTERING TYPE 2 (ONKORZHENSKYY) USED IF AVAILABLE
SHORT DECELERATION LENGTH = 0.000000 COLLISIONS.

THERMAL MOTION OF GAS INCLUDED
ELECTRIC FIELD = 1000.0000 VOLTS/CM.
MAGNETIC FIELD = 0.0000 KILOGAUSS.
ANGLE BETWEEN ELECTRIC AND MAGNETIC FIELD = 0.000 DEGREES.
CYCLOTRON FREQ. = 0.00000000 RAD/MS/PIECOSECOND
INITIAL ELECTRON ENERGY = 0.000 EV.
TOTAL NUMBER OF REAL COLLISIONS = 2000000000
WELL COLLISION FREQUENCY FOR EACH GAS COMPONENT IN UNITS OF (1/10**12/SEC)
0.1470+01 0.1100+01 0.3070+01
0.0000+00 0.0000+00 0.0000+00

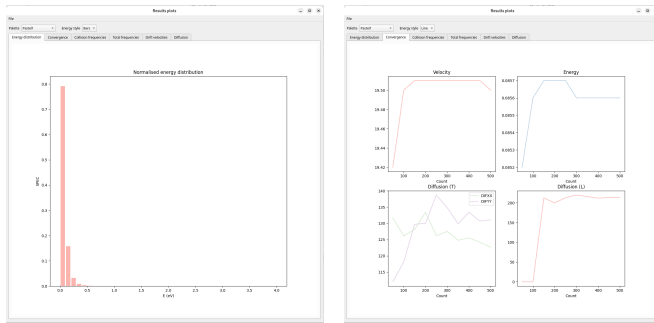
VEL POS TIME ENERGY COUNT DIFFX DIFFY DIFFZ
19.42 0.2240+00 0.1120+00 0.8852 50 131.7 112.0 0.0
```

From Text Output → Structured Data

- ▶ Canonical RunResult model
- ▶ Extracted:
 - ▶ Transport coefficients
 - ▶ Collision frequencies
 - ▶ Convergence tables
 - ▶ Energy distributions
- ▶ Export formats:
 - ▶ CSV
 - ▶ JSON
 - ▶ XML
- ▶ Reproducible analysis pipelines

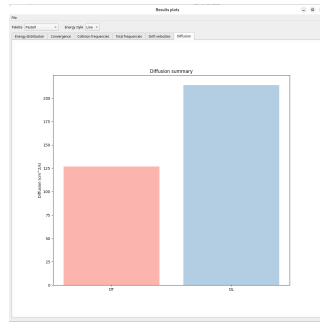
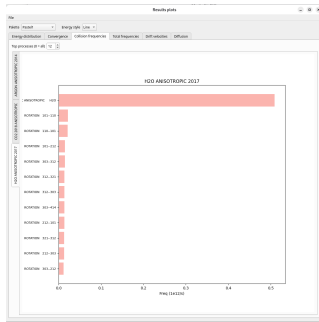


Built-in Plotting Tools



Immediate visual feedback without external scripts.

Built-in Plotting Tools (cont.)



Immediate visual feedback without external scripts.

Project repository

- ▶ GitLab: <https://gitlab.com/micrenda/magboltz-gui/>
- ▶ License: MIT
- ▶ Documentation: README in the repository

Setup: pip (bundled Qt)

Recommended when you want Qt bundled with the app.

- ▶ Python environment
- ▶ Magboltz binary available in PATH

Commands:

```
pip install "magboltz-gui[qt]"  
magboltz-gui
```

Setup: pip (system Qt)

Recommended on Linux when using system PyQt6.

- ▶ Install system Qt bindings
- ▶ Magboltz binary available in PATH

Commands (Debian/Ubuntu):

```
sudo apt install python3-pyqt6  
pip install magboltz-gui  
magboltz-gui
```

Stephen F. Biagi

(1949–2025)

We dedicate this work to his memory.

His pioneering contributions to modeling
electron transport in gases profoundly shaped
detector physics.

This interface is a modest tribute to his legacy.